

```
!git clone https://github.com/Nishviprp/CodeGen-LLM.git
%cd CodeGen-LLM/project
!pip install -q transformers peft accelerate datasets rouge-score sentencepiece
```

```
Cloning into 'CodeGen-LLM'...
remote: Enumerating objects: 46, done.
remote: Counting objects: 100% (46/46), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 46 (delta 9), reused 22 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (46/46), 11.63 MiB | 5.45 MiB/s, done.
Resolving deltas: 100% (9/9), done.
[Errno 2] No such file or directory: 'CodeGen-LLM/project'
/content
  Preparing metadata (setup.py) ... done
  Building wheel for rouge-score (setup.py) ... done
```

```
!python data/prepare_data.py
```

```
python3: can't open file '/content/data/prepare_data.py': [Errno 2] No such file
```

```
%cd /content/CodeGen-LLM
```

```
/content/CodeGen-LLM
```

```
!python data/prepare_data.py
```

```
Downloading CodeParrot dataset (5000 examples)...
This may take a few minutes on first run (downloads ~500MB).

README.md: 100% 1.30k/1.30k [00:00<00:00, 3.31MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF
Resolving data files: 100% 54/54 [00:00<00:00, 42469.98it/s]
  Collected 500/5000 examples...
  Collected 1000/5000 examples...
  Collected 1500/5000 examples...
  Collected 2000/5000 examples...
  Collected 2500/5000 examples...
  Collected 3000/5000 examples...
  Collected 3500/5000 examples...
  Collected 4000/5000 examples...
  Collected 4500/5000 examples...
  Collected 5000/5000 examples...

Done. Saved 5000 training examples to data/python_dataset.json
(Skipped 0 files that were too short)

Downloading HumanEval benchmark (164 tasks)...
README.md: 100% 6.52k/6.52k [00:00<00:00, 18.0MB/s]
Traceback (most recent call last):
  File "/content/CodeGen-LLM/data/prepare_data.py", line 137, in <module>
```

```

prepare_eval_data(
File "/content/CodeGen-LLM/data/prepare_data.py", line 112, in prepare_eval_data
    dataset = load_dataset("openai_humaneval", split="test")
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/datasets/load.py", line 1392, in
    builder_instance = load_dataset_builder(
                       ^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/datasets/load.py", line 1132, in
    dataset_module = dataset_module_factory(
                     ^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/datasets/load.py", line 1031, in
    raise e1 from None
File "/usr/local/lib/python3.12/dist-packages/datasets/load.py", line 1004, in
    ).get_module()
      ^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/datasets/load.py", line 591, in
    standalone_yaml_path = cached_path(
                           ^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/datasets/utils/file_utils.py", line 1
    ).resolve_path(url_or_filename)
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/huggingface_hub/hf_file_system.py", line 1
    parsed = parse_hf_uri(f"{constants.HF_PROTOCOL}{path}")
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_hf_url_utils.py", line 1
    return _parse_repo_body(location, type_, raw=raw)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_hf_url_utils.py", line 1
    raise HfUriError(uri=raw, msg=f"Repository id must be 'namespace/name', got '{raw}'")
huggingface_hub.errors.HfUriError: Invalid HF URI 'hf://datasets/openai_humaneval'

```

```
!python models/finetune.py
```

```

Using device: cuda
Loading base model: Salesforce/codegen-350M-mono
config.json: 100% 999/999 [00:00<00:00, 6.28MB/s]
tokenizer_config.json: 100% 240/240 [00:00<00:00, 1.53MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF token.
vocab.json: 100% 798k/798k [00:00<00:00, 92.7MB/s]
merges.txt: 100% 456k/456k [00:00<00:00, 106MB/s]
added_tokens.json: 100% 1.00k/1.00k [00:00<00:00, 2.99MB/s]
special_tokens_map.json: 100% 90.0/90.0 [00:00<00:00, 654kB/s]
tokenizer.json: 100% 2.11M/2.11M [00:00<00:00, 150MB/s]
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
pytorch_model.bin: 100% 797M/797M [00:11<00:00, 69.7MB/s]
Loading weights: 100% 165/165 [00:00<00:00, 35213.97it/s]
[transformers] CodeGenForCausalLM LOAD REPORT from: Salesforce/codegen-350M-mono
Key | Status |
-----+-----+
transformer.h.{0...19}.attn.causal_mask | UNEXPECTED |

```

Notes:

- **UNEXPECTED:** can be ignored when loading from different task/architecture; no

Traceback (most recent call last):

```
File "/content/CodeGen-LLM/models/finetune.py", line 190, in <module>
```



```
Collected 4500/5000 examples...
Collected 5000/5000 examples...
```

```
Done. Saved 5000 training examples to data/python_dataset.json
(Skipped 0 files that were too short)
```

```
Downloading HumanEval benchmark (164 tasks)...
README.md: 100% 6.52k/6.52k [00:00<00:00, 17.8MB/s]
openai_humaneval/test-00000-of-00001.par(...): 100% 83.9k/83.9k [00:01<00:00, 80.9
Generating test split: 100% 164/164 [00:00<00:00, 6923.04 examples/s]
Done. Saved 164 HumanEval tasks to data/human_eval.json
```

```
All data ready. Next step: python models/finetune.py
```

```
!pip uninstall -y torchao
!pip install -q "peft==0.11.1"
```

```
Found existing installation: torchao 0.10.0
Uninstalling torchao-0.10.0:
  Successfully uninstalled torchao-0.10.0
  _____ 251.6/251.6 kB 20.4 MB/s eta 0:00:00
```

```
!python models/finetune.py
```

```
Using device: cuda
Loading base model: Salesforce/codegen-350M-mono
Warning: You are sending unauthenticated requests to the HF Hub. Please set a ...
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100% 165/165 [00:00<00:00, 24013.19it/s]
[transformers] CodeGenForCausalLM LOAD REPORT from: Salesforce/codegen-350M-mono
Key | Status | |
-----+-----+--
transformer.h.{0...19}.attn.causal_mask | UNEXPECTED | |
```

```
Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; no
trainable params: 983,040 || all params: 357,695,488 || trainable%: 0.2748
Loading training data from data/python_dataset.json...
  Loaded 5000 training examples
```

```
Map: 100% 5000/5000 [00:30<00:00, 162.07 examples/s]
Tokenized 5000 examples
```

```
[transformers] warmup_ratio is deprecated and will be removed in v5.2. Use `warm
```

```
Starting fine-tuning...
```

```
0% 0/939 [00:00<?, ?it/s]Traceback (most recent call last):
  File "/content/CodeGen-LLM/models/finetune.py", line 190, in <module>
    finetune()
  File "/content/CodeGen-LLM/models/finetune.py", line 180, in finetune
    trainer.train()
  File "/usr/local/lib/python3.12/dist-packages/transformers/trainer.py", line 1
```



```
{'loss': '0.9312', 'grad_norm': '0.1484', 'learning_rate': '1.154e-05', 'epoch':  
{'loss': '0.9057', 'grad_norm': '0.1426', 'learning_rate': '9.789e-06', 'epoch':  
{'loss': '0.9253', 'grad_norm': '0.1709', 'learning_rate': '8.04e-06', 'epoch':  
{'loss': '0.92', 'grad_norm': '0.2012', 'learning_rate': '6.353e-06', 'epoch': '  
{'loss': '0.9034', 'grad_norm': '0.1523', 'learning_rate': '4.778e-06', 'epoch':  
{'loss': '0.8833', 'grad_norm': '0.1699', 'learning_rate': '3.365e-06', 'epoch':  
{'loss': '0.9213', 'grad_norm': '0.1572', 'learning_rate': '2.157e-06', 'epoch':  
{'loss': '0.9342', 'grad_norm': '0.1475', 'learning_rate': '1.191e-06', 'epoch':  
{'loss': '0.9058', 'grad_norm': '0.1729', 'learning_rate': '4.982e-07', 'epoch':  
{'loss': '0.9008', 'grad_norm': '0.1641', 'learning_rate': '9.907e-08', 'epoch':  
{'train_runtime': '6003', 'train_samples_per_second': '2.499', 'train_steps_per_  
100% 939/939 [1:40:03<00:00, 6.39s/it]
```

Adapter saved to models/finetuned-adapter/
Next step: python evaluation/evaluator.py

```
import shutil  
shutil.make_archive('finetuned-adapter', 'zip', 'models/finetuned-adapter')  
from google.colab import files  
files.download('finetuned-adapter.zip')
```

Start coding or [generate](#) with AI.

